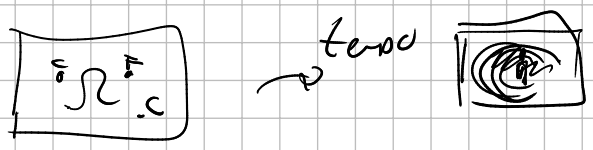
EQUAZIONI PARABOLICHE

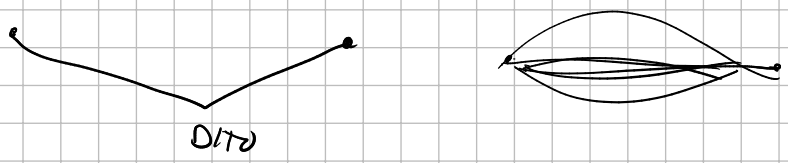
$$\partial_t u - a\,\partial_{xx} u = f(t,x) \qquad a>0$$

1D SPAZIALE x
1 TEMPORALE t } 2 DIMENSIONI

- CONDUZIONE CALORE

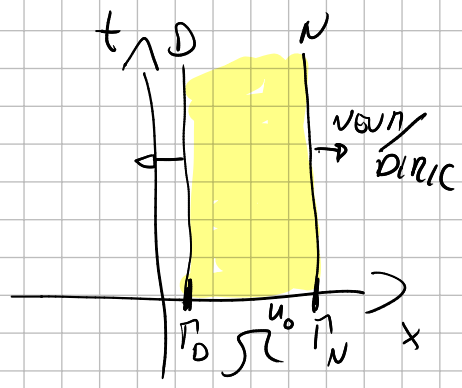


- ELASTICITÀ



- PROBLEMA DI CAUCHY

$$\begin{cases} \text{PDE} & \partial_t u - a\,\partial_{xx} u = f \qquad t>0,\ x\in\Omega\ (\subset\mathbb{R}) \\ \text{IC} & u(t=0,x) = u_0(x) \qquad \forall x\in\Omega \\ \text{BC. DIR} & u(t,x) = u_D(t,x) \qquad \forall x\in\Gamma_D\subset\partial\Omega,\ \forall t>0 \\ \text{BC. NEUM} & u_x(t,x)\cdot n_x = u_N(t,x) \qquad \forall x\in\Gamma_N\subset\partial\Omega,\ \forall t>0 \end{cases}$$

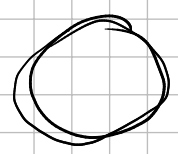


IC = INITIAL CONDITION
BC = BOUNDARY CONDITIONS

- BC PERIODICHE (IN ALTERNATIVA DIRICHLET/NEUMANN)

$$u(t,a) = u(t,b) \qquad \forall t>0 \qquad \Omega = [a,b]$$

Ω = CIRCONFERENZA



- FOURIER $\quad e^{ikx}$

$i = \sqrt{-1}$

$k \in \mathbb{Z} \qquad x \in \Omega = [0, 2\pi]$

- $\underbrace{\partial_x}_{} \underbrace{e^{ikx}}_{\text{AUTO FUNZIONE}} = \overbrace{ik}\ \underbrace{e^{ikx}}_{\text{AUTO FUNZIONE}}$

$\partial_{xx} e^{ikx} = -k^2 e^{ikx}$

$\underbrace{A}_{\text{DERIVATA}} \cdot V = \underbrace{\lambda}_{\text{AUTO VALORE}} \cdot V$

$$e^{ikx} = \cos(kx) + i \sin(kx)$$

$$\cos(kx) = \frac{e^{ikx} + e^{-ikx}}{2} \qquad \sin(kx) = \frac{e^{ikx} - e^{-ikx}}{2i}$$

$$\partial_{xx} \cos(kx) = -k^2 \cos(kx)$$

FUNZIONE DI BASE $\{ e^{ikx} \}_{k \in \mathbb{Z}}$ $\qquad \Omega = [-\pi, \pi]$

$$u_0(x) = \sum_{k \in \mathbb{Z}} c_k e^{ikx} \qquad c_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} u_0(x) e^{-ikx} \, dx$$

$c_k \in \mathbb{C}$

$u_0 \to c_k$ TRASFORMATA DI FOURIER

$c_k \to u_0$ ANTI TRASFORMATA

$u_0 : \mathbb{R} \to \mathbb{R}$

$c : \mathbb{Z} \to \mathbb{C}$

- TEOREMA DI PARSEVAL

$$\| c \|_2^2 = \sum_{k \in \mathbb{Z}} |c_k|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |u_0(x)|^2 dx$$

$$= \frac{1}{2\pi} \| u_0 \|_2^2 < \infty$$

$$\begin{cases} \partial_t u(x,t) - a\,\partial_{xx} u(t,x) = 0 & \forall\ x \in [-\pi,\pi]\ \ t>0 \\ u(t=0,x) = u_0(x) & \forall\ x \in [-\pi,\pi] \\ u(t,-\pi) = u(t,\pi) & \forall\ t>0 \end{cases}$$

IC

BC. PER

$$u_0(x) = \sum_{k\in\mathbb{Z}} c_k(0)\cdot e^{ikx}$$

$$u(x,t) = \sum_{k\in\mathbb{Z}} \underbrace{c_k(t)}_{\uparrow}\cdot e^{ikx}$$

SUPPONIAMO CHE LA NOSTRA SOLUZIONE SI POSSA SCRIVERE COSÌ

INVECE DI TROVARE $u(x,t) \Rightarrow$ CERCHIAMO $c_k(t)$

$$\sum_{k\in\mathbb{Z}} \partial_t c_k(t)\cdot \underbrace{e^{ikx}} - a\sum_{k\in\mathbb{Z}} c_k(t)\cdot(-k^2)\underbrace{e^{ikx}} = 0$$

e^{ikx} è UNA BASE ORTONORMALE $\langle f,g\rangle = \frac{1}{2\pi}\int f\cdot\bar{g}\,dx$

$\bar{g} = \overline{(a+ib)} = a-ib$

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-izx}\left(\text{equazione}\right) = 0 \Rightarrow$$ TUTTI I TERMINI $k=z$ SOPRAVVIVONO

$k\neq z$ SPARISCONO

$$\Rightarrow \forall k\in\mathbb{Z} \quad \boxed{\partial_t c_k(t) + ak^2 c_k(t) = 0}$$ ODE

$$c_k(t) = e^{-ak^2\cdot t}\cdot c_k(0) \qquad \forall k\in\mathbb{Z}$$

$\Downarrow$

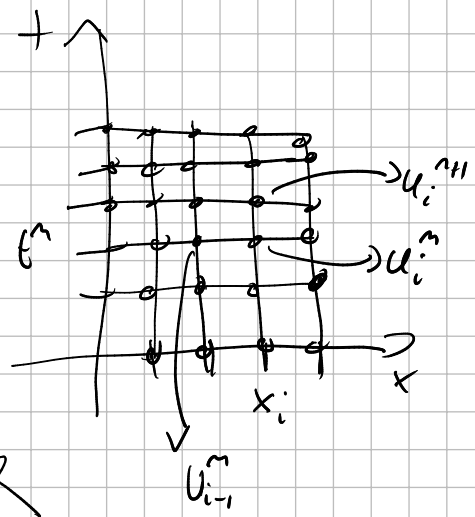
ABBIAMO LA NOSTRA SOLUZIONE $\quad u(t,x) = \sum_{k\in\mathbb{Z}} c_k(t)\cdot e^{ikx}$

DIFFERENZE FINITE $\times$ PARABOLICI

$\Omega = [a,b] \qquad a = x_0 < \dots < x_i < \quad < x_{N_x} = b$

$0 = t^0 < \dots t^n < \quad < t^{N_t} = T$

$$(\partial_{xx} u)_i \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2}$$



- EULERO ESPLICITO

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} - \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2} = 0$$

$\forall i = 1, \dots, N_x - 1 \qquad \forall n = 0, \dots, N_t - 1$

NO MATRICI

NO SISTEMA LINEARE

- EULERO IMPLICITO

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} - \underbrace{\frac{u_{i+1}^{n+1} - 2u_i^{n+1} + u_{i-1}^{n+1}}{\Delta x^2}}_{\text{IMPLICITO} \rightarrow \text{SISTEMA LINEARE}} = 0 \qquad (1^\circ \text{ ORDINE})$$

- RUNGE KUTTA ORDINE > 1

es. CRANK-NICOLSON MEDIA E.E e E.I.

IMPLICITI $\rightarrow$ SYS LINEARE TRIDIAGONALE (ORDINE 2 IN SPAZIO)

ORDINE DI ACCURATEZZA? (CONSISTENZA)

EULERO ESPLICITO

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} - \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2} = 0$$

$$e^{EE}_{\Delta t, \Delta x} = \frac{u(x_i, t^{n+1}) - u(x_i, t^n)}{\Delta t} - \frac{u(x_{i+1}, t^n) - 2u(x_i, t^n) + u(x_{i-1}, t^n)}{\Delta x^2}$$

ERRORE DI CONSISTENZA: INSERISCO LA SOLUZIONE ESATTA IN METODO e TAYLOR EXPANSION.

TAYLOR IN (x_i, t^n)

$$e^{\sigma\tau}_{\Delta t, \Delta x} = \frac{\left(\cancel{u(x_i,t^n)} + \Delta t\, \partial_t u(x_i,t^n) + O(\Delta t^2)\right) - \cancel{u(x_i,t^n)}}{\Delta t}$$

$$- \frac{\left\{ \left(\cancel{u(x_i,t^n)} + \cancel{\Delta x \cdot \partial_x u(x_i,t^n)} + \frac{\Delta x^2}{2} \partial_{xx} u(x_i,t^n) + \cancel{\frac{\Delta x^3}{6} \partial_{xxx} u(x_i,t^n)} + O(\Delta x^4) \right) - \cancel{2u(x_i,t^n)} + \left(\cancel{u(x_i,t^n)} - \cancel{\Delta x\, \partial_x u(x_i,t^n)} + \frac{\Delta x^2}{2} \partial_{xx} u(x_i,t^n) - \cancel{\frac{\Delta x^3}{6} \partial_{xxx} u} + O(\Delta x^4) \right) \right.}{\Delta x^2}$$

$$= \partial_t u(x_i,t^n) + O(\Delta t) - \partial_{xx} u(x_i,t^n) + O(\Delta x^2)$$

$$= O(\Delta t) + O(\Delta x^2).$$

errore del PRIM'ORDINE IN TEMPO

→ SECOND'ORDINE IN SPAZIO

$\Delta t \sim \Delta x$.

SEMI DISCRETIZZAZIONE (IN SPAZIO) / METHOD OF LINES

$$\partial_t u_i(t) = \frac{u_{i+1}(t) - 2u_i(t) + u_{i-1}(t)}{\Delta x^2} \qquad \underline{u} = \begin{pmatrix} u_0 \\ u_1 \\ \vdots \\ u_{N_x} \end{pmatrix}(t)$$

$$\boxed{\partial_t \underline{u}(t) = \underline{\underline{A}} \cdot \underline{u}(t)} + \underbrace{g(t)}$$

QUALCOSA CHE NON DIPENDE DA U

es. FORZANTE

SISTEMA DI ODE

$$A = \text{tri diag}\begin{pmatrix} -\frac{2}{\Delta x^2} & \frac{1}{\Delta x^2} & 0 \\ \frac{1}{\Delta x^2} & \ddots & \ddots \\ 0 & \ddots & \ddots \end{pmatrix}$$

• REGIONI DI STABILITÀ X RUNGE-KUTTA

(SISTEMI LINEARI)

$$\partial_t u = A u$$

DIAGONALIZZARE

$$Z^T A \underbrace{Z}_{\text{AUTOVETT}} = \underbrace{D}_{\text{AUTOVAL}} \qquad Z^T Z = I$$

$$\partial_t u = Z D Z^T u$$

$$\partial_t \underbrace{Z^T u}_{y} = D \underbrace{Z^T u}_{y}$$

$$\partial_t y = D y$$

$\rightarrow \partial_t y_i = \lambda_i y_i$ OGNI EQUAZIONE È DISACCOPPIATA

EQUAZIONE DI DAHLQUIST $\qquad \lambda_i \in \mathbb{C}$

STABILITÀ?

$$y^{n+1} = \underbrace{R(\lambda_i \Delta t)}_{\text{METODO RK}} \cdot y^n$$

es. $y^{n+1} = (1 + \Delta t \lambda)\, y^n$ EULERO ESPLICITO

es. $y^{n+1} - \lambda \Delta t\, y^{n+1} = y^n$ EULERO IMPLICITO

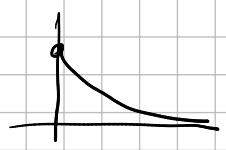
$$y^{n+1}(1 - \lambda \Delta t) = y^n \qquad y^{n+1} = \frac{1}{1 - \lambda \Delta t} y^n$$

$y' = \lambda y \qquad \Delta t \cdot \lambda = z \in \mathbb{C} \qquad Re(\lambda) < 0$

(IN DISSIPANTE)

$$y^{n+1} = R(z) \cdot y^n$$

$$y = e^{\lambda t}\, y(0)$$



$$|y^{n+1}| \le |y^n|$$

$$\| $$

$$|R(z)\cdot y^n| \le |R(z)|\cdot|y^n|$$

Se $|R(z)| \le 1$ CON $z \in \mathbb{R}$ $\Rightarrow$ STABILE

REGIONE DI STABILITÀ $\qquad y' = \lambda y$

$$S = \{ z \in \mathbb{C} : \underbrace{|R(z)|} \le 1 \} \qquad y^{n+1} = R(z)\cdot y^n$$

es. EULERO ESPLICITO $\qquad y^{n+1} = y^n + \overbrace{\Delta t \lambda}^{z}\, y^n$

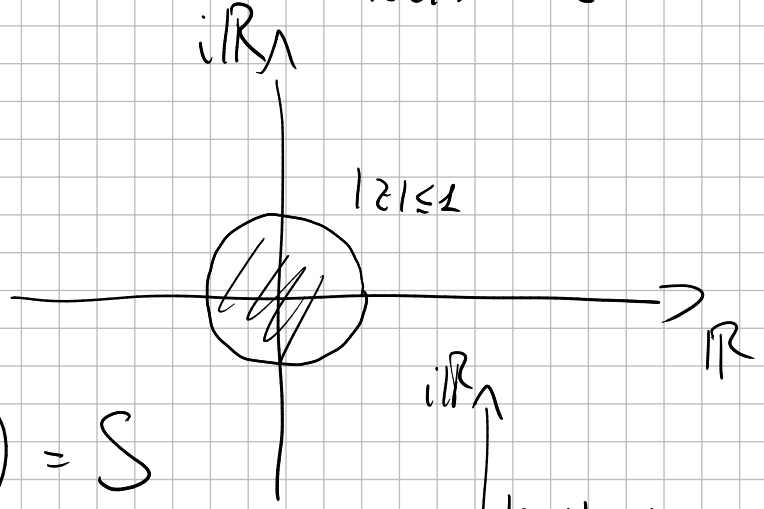
$$= (1+z)\cdot y^n$$

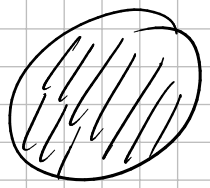
$$R(z) = 1+z$$

$$|R(z)| \le 1$$

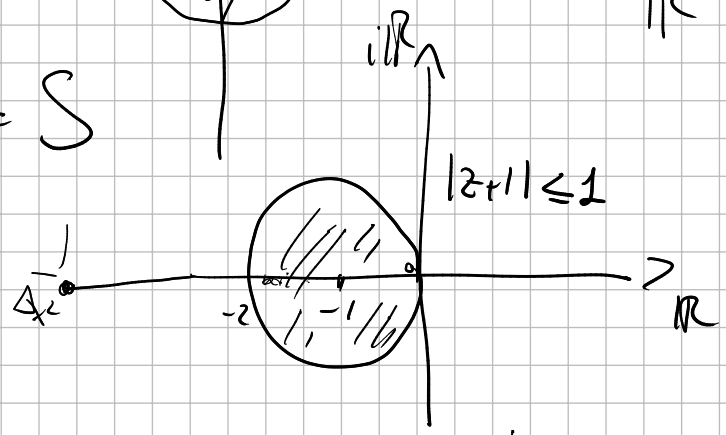
$$\underbrace{|1+z|} \le 1$$

$$|z| \le 1$$



Se $\lambda \cdot \Delta t \in$  $= S$

$$\Rightarrow |y^{n+1}| \le |y^n|$$



$$\frac{1}{\Delta x^2}\left(\cancel{\begin{matrix} 8 & -1 \\ -1 & \end{matrix}}\right)$$

$$|z| = |\Delta t \cdot \lambda| \sim \left|\frac{1}{\Delta x^2}\cdot \Delta t\right| \le 2$$

$$\Delta t \sim \Delta x^2$$

PER EULERO ESPLICITO

$\Delta t \sim \Delta x^2$ STABILE

es. EULERO IMPLICITO

$$y^{n+1} = y^n + \underbrace{\lambda \Delta t}_{z} \cdot y^{n+1}$$

$$(1-z)\, y^{n+1} = y^n$$

$$y^{n+1} = \frac{1}{1-z} \cdot y^n$$

$$R(z) = \frac{1}{1-z}$$

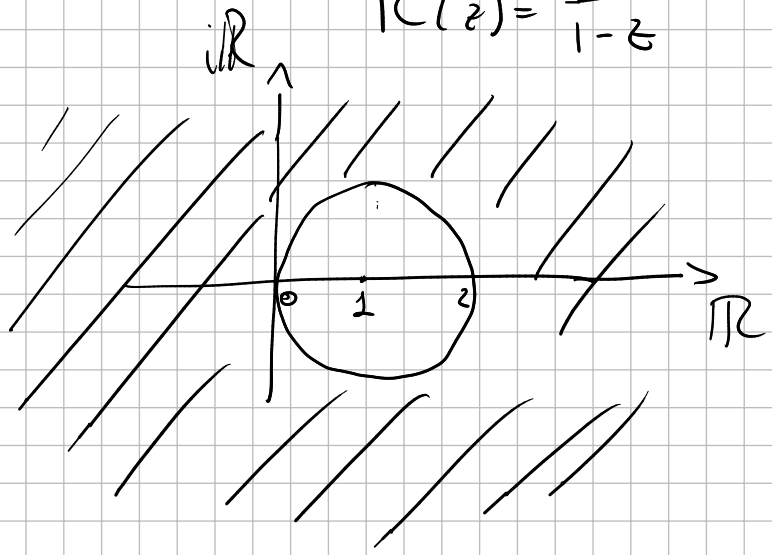
$$\left| \frac{1}{1-z} \right| \leq 1$$

$$\Updownarrow$$

$$\frac{1}{|1-z|} \leq 1$$

$$\Updownarrow$$

$$|1-z| \geq 1$$

$\forall z \in \mathbb{C}^-$

STABILE!